



Question: What behavior is least likely to cause a traffic incident? Choose from the following answers only.

A) Slow down.

B) Move right.

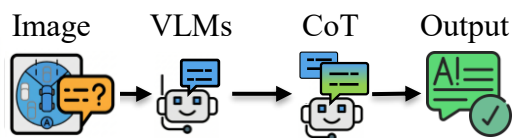
C) Speed up.

D) Maintain speed.

E) Move left.

F) None of the options.

(a) Standard SFT

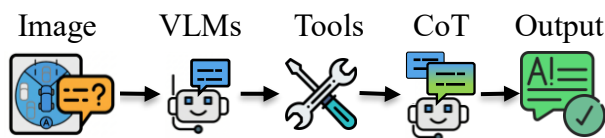


<think>1. In the front view... **there is a pedestrian crossing the road.** This indicates a potential hazard...
3. Moving right or left could be risky due to the presence of parked cars...
5. Maintaining speed is not ideal...
Therefore, slowing down is prudent. </think>



<answer>
Final Answer: A) Slow down.
</answer>

(b) Tool-Augmented



<tool>function:"open_vocabulary_detector",output:"Pedestrians detected nearby."</tool>



<think>Pedestrians are nearby indicates a potential hazard. Then I must estimate distances to ensure safety.</think>

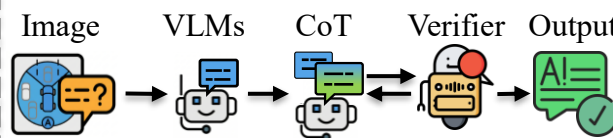


<tool>function:"depth_estimator",output:"Pedestrian distances cannot be determined."</tool>



<answer>
Final Answer: A) Slow down.
</answer>

(c) Critique-Driven RL (Teacher)



<think> 1. Front images show a pedestrian crossing... 3. ...slowing down is the safest to prevent incident. ... </think>



Critique:... You chose to slow down, which is conservative... Re-examine the scenario: Is there a reason to slow down?

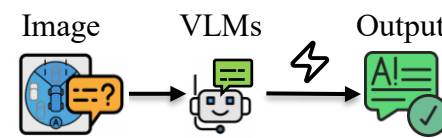


<think>... 3. Feedback suggests slowing down is..., as it could block traffic. 4. ... Maintaining speed keeps the vehicle clear... </think>



<answer>
Final Answer: D) Maintain speed.
</answer>

(d) Latent Thought Distillation (Student)



<answer>
Final Answer: D) Maintain speed.
</answer>

Standard SFT

✗ **Low Accuracy**
✗ **Hallucination**

Tool-Augmented

✓ **Higher Accuracy**
✓ **Tool-Augmented CoT**
✗ **Massive Tokens**

Ours-Teacher

✓ **SOTA Accuracy**
✓ **Reasoning Consistency**
✓ **Less Hallucinations**

Ours-Student

✓ **High Accuracy**
✓ **Low Token**